

Study Design – “Social Battery Flower”

The goal of the study is to evaluate how users perceive an interactive physical artifact designed to communicate social availability and encourage conscious disconnection from digital devices. The study focuses on whether users understand the concept, emotionally connect with it, and perceive it as socially acceptable and supportive in everyday life.

Based on the feedback, the study should focus on a smaller number of clearly defined dimensions instead of covering too many topics at once.

Refined Research Focus

Research Question

How do users perceive a physical interactive flower artifact that visualizes social availability and supports reduced digital dependency?

Focused Evaluation Dimensions

To avoid spreading the study too broadly, the evaluation will focus on three main dimensions:

1. Intuitiveness & Understandability

- ◆ Do users understand the meaning of the flower states and interactions?
- ◆ Is the concept self-explanatory?

2. Emotional & Social Acceptance

- ◆ Do users perceive the system as comforting, supportive, or socially appropriate?
- ◆ Would they feel comfortable using it in real life?

3. Perceived Support for Digital Disconnection

- ◆ Do users believe the artifact could help reduce phone usage or digital overload?
- ◆ Does the physical interaction feel less stressful than digital communication?

These dimensions align strongly with the project goals and are realistic to evaluate within a short pilot study.

Hypotheses

- H1. Participants will perceive the flower interaction as intuitive and easy to understand.
- H2. Participants will perceive the flower as an emotionally supportive and socially acceptable way to communicate social availability.
- H3. Participants will perceive the flower as encouraging conscious disconnection from digital devices.

Study Type

The study will use a **mixed-methods approach**:

- ◆ Quantitative questionnaire (Likert-scale ratings)
- ◆ Qualitative semi-structured interview

Because the prototype may not be fully functional in time, the study is designed to work with:

- ◆ physical prototype,
- ◆ concept images,
- ◆ scenario descriptions,
- ◆ or a short live demonstration.

A video is therefore optional rather than required.

Participants

Target Group & Sample Size

Young adults and students who frequently use smartphones and digital communication platforms.

Qualitative Pilot Study

- ◆ 5 - 8 participants
- ◆ In-person sessions
- ◆ Think-aloud interaction + interview

Optional Online Survey Extension

- ◆ 15 - 25 participants
- ◆ Image-based concept presentation + questionnaire only

Study Setup

Materials

- ◆ Physical flower prototype (if available)
- ◆ Printed concept images/renderings
- ◆ Short usage scenario descriptions
- ◆ Questionnaire (digital or paper)
- ◆ Interview guide and Consent form

Study Procedure

1. Introduction (2-3 min)

Participants receive a short introduction to the project, an explanation of the study purpose, and a consent information. Example:

“This study investigates how people perceive a physical artifact that represents social availability and encourages reduced digital dependency.”

2. Concept Presentation / Demonstration (3-5 min)

Depending on prototype readiness:

Option A – Working Prototype

Participants interact with the flower themselves.

Option B – Partial Prototype

We will demonstrate the flower states, interaction behavior, availability changes, phone-locking concept, and a contextual adaptation.

Option C – No Functional Prototype

Participants receive scenario descriptions, printed visuals, and a storyboard-style walkthrough. This flexibility ensures the study can still be conducted successfully.

For an **Example Usage Scenario** participants are asked to imagine:

> “You are studying at home and feel socially exhausted after a long day. The flower slowly closes its petals to indicate low social availability. Your friends can understand your current social state without needing constant messaging or online status updates. The system also gently encourages reduced phone checking.”

3. Questionnaire (5-7 min)

The questionnaire focuses only on the three core dimensions using a 7-point Likert scale (1 = Strongly disagree, 7 = Strongly agree):

A. Intuitiveness & Understanding

1. I understood the purpose of the flower immediately.
2. The flower states were easy to interpret.
3. The interaction with the system felt intuitive.
4. I understood how the system communicates social availability.

B. Emotional & Social Acceptance

5. The flower feels calming and non-intrusive.
6. I would feel comfortable using this system in everyday life.

7. The concept feels emotionally supportive.
8. I would prefer this type of communication over constantly messaging others about my availability.

C. Support for Digital Disconnection

9. The system encourages more conscious phone usage.
10. The concept could help reduce digital overload.
11. The physical representation feels less stressful than digital notifications.
12. The system could help create healthier social boundaries.

Optional Standardized Questionnaire Integration

To address the feedback regarding standardized scales, we can integrate short adapted items from the **Technology Acceptance Model (TAM)**, **System Usability Scale (SUS)** or **Trust in Automation** scales.

For a Pilot study we do not need to include full questionnaires, but a small adapted subset is sufficiently. The subset includes *perceived usefulness, ease of use, trust, and willingness to use*.

4. Semi-Structured Interview (5-10 min)

The interview explores experiences in more depth and should follow a consistent structure. The Interview Questions are as followed:

Understanding & Interpretation

1. What do you think the flower communicates?
2. Which interaction or flower state was most understandable to you?
3. Was anything confusing or unclear?

Emotional Perception

4. How did the artifact make you feel?
5. Did the concept feel comforting, supportive, or stressful in any way?
6. Would you feel comfortable displaying your social availability through this object?

Digital Disconnection

7. Do you think this system could realistically help reduce phone usage?
8. In which situations would you personally use such a system?
9. Do you think a physical artifact changes the experience compared to an app or phone notification?

Reflection & Improvement

10. What did you like most about the concept?
11. What would you improve or change?
12. Would you use such a system in your own environment? Why or why not?

Data Collection

Quantitative Data

- ◆ Likert-scale ratings
- ◆ Mean scores per dimension
- ◆ Descriptive statistics

Qualitative Data

- ◆ Interview notes/audio recordings
- ◆ Thematic clustering:
 - * understanding,
 - * emotional response,
 - * social comfort,
 - * perceived usefulness,
 - * concerns/improvements.

Data Analysis

Quantitative

- ◆ Average ratings per dimension
- ◆ Compare strongest and weakest aspects
- ◆ Identify general acceptance trends

Qualitative

We will use lightweight thematic analysis: recurring themes, concerns, positive reactions, misunderstandings, suggestions.

Because this is a pilot study, deep statistical analysis is not necessary.

Expected Outcomes

The study aims to investigate whether:

- ◆ users intuitively understand the flower metaphor,

- ◆ the physical artifact creates a calmer social interaction experience,
- ◆ users perceive the concept as a meaningful alternative to digital communication overload.

Practical Recommendations for the Timeline

Since the study takes place very soon:

Prioritize

- ✓ a clear explanation of the concept,
- ✓ consistent demonstration,
- ✓ strong questionnaire design,
- ✓ structured interview questions.

Do NOT spend too much time on

- X polishing every prototype interaction,
- X producing a fully edited video,
- X implementing advanced adaptive behaviors.

A clear prototype concept and a well-structured study are more important than technical completeness for this pilot evaluation.